ASSIGNMENT 1 — Q2

Multi-Type Data Summary

JavaScript solution

```
// Q2: Multi-Type Data Summary
"use strict";

const userName = "Mohit";
const age = 20;
const isStudent = true;
const skills = ["JavaScript", "HTML", "CSS"];
const profile = { city: "Agra", course: "B.Tech CSE" };
const middleName = null;
let optionalValue;

const rows = [
    { Label: "userName", Value: userName, Type: typeof userName },
    { Label: "age", Value: age, Type: typeof age },
    { Label: "isStudent", Value: isStudent, Type: typeof isStudent },
    { Label: "skills", Value: skills.join(", "), Type: Array.isArray(skills) ? "array" : typeof skills },
    { Label: "profile", Value: JSON.stringify(profile), Type: typeof profile },
    { Label: "middleName", Value: String(middleName), Type: middleName === null ? "null" : typeof middleName },
    { Label: "optionalValue", Value: String(optionalValue), Type: typeof optionalValue }
];

console.table(rows);
```

Logic: Uses typeof for primitive/object values and Array.isArray() to label arrays correctly; null is specially identified because typeof null is 'object'.

How to test: Save this code as a .js file and run it with Node.js using `node filename.js`, or paste it into the browser console.